GROUP 16

NAME	REGISTRATION NUMBER
LUUTU RAHMA NAMAYANJA	25/U/03438/PS
NAMBUYA RAUDHA	25/U/03516/PS
OJAMBO NICHOLAS	25/U/03551/PS
MUMBERE BREILINE	25/U/03472/PS
LUUTU GREGORY VALENTINO	25/U/0347

DATABASE REQUIREMENTS

1. **System Users and Access Roles**: In order to ensure operational security, the database will support four primary user roles which are;
    - Pharmacy Manager (Admin): Manages overall system users, reviews the business analytics and monitors high-level financial and inventory reports.
    - Inventory Manager: Handles new stock deliveries, logs stock details and generates the batch codes.
    - Pharmacists: Validate the patients' prescriptions, manage dispensing and ensure safe medication distribution.
    - Cashiers: Process customer payments which are both cash and mobile money and also confirm that the customer got their medicine.

2. CORE DATABASE ENTITIES AND ATTRIBUTES

- **USERS**; stores the staff details.
    - UserID (Primary Key)
    - CategoryID (Foreign key)
    - FullName
    - UserName
    - Password
    - Role

- **DRUGS; holds the general medications into logical groups**
    - DrugID (Primary key)
    - GenericName
    - BrandName
    - UnitMeasure
    - UnitPrice

- **INVENTORY_BATCHES; table for tracking stock, pricing variations per batch and expiries**
    - BatchID (Primary key)
    - DrugID (Foreign key)
    - SupplierID (Foreign key)
    - Batchnumber
    - Quantityreceived
    - Quantityleft
    - MFD

- Expirydate
- Unitprice

- **TRANSACTIONS; records check-out meta-data**
  - TransactionID (Primary key)
  - UserID (Foreign key)
  - Timestamp
  - Totalamount
  - Paymentmethod (Cash/Mobile money)

- **SALE_ITEMS; junction table connecting sales to exact drug batches sold.**
  - SalesItemID (Primary key)
  - TransactionID (Foreign key)
  - BatchID (Foreign key)
  - Quantitysold
  - Subtotal

- **CATGORIES; Organizes general medications into logical classification groups.**
  - CategoryID (Primary key)
  - CategoryName
  - Description

- **SUPPLIERS; Tracks pharmaceutical vendors and manufacturers who deliver inventory batches**
  - SupplierID
  - SupplierName
  - ContactPerson
  - PhoneNumber
  - Email
  - Address

3. KEY RELATIONSHIPS AND BUSINESS LOGIC

- **Drugs to Inventory Batches (1:M, WEAK RELATIONSHIP)**: One generic drug record in the DRUGS table can be linked to multiple stock entries in the INVENTORY_BATCHES table with distinct batch numbers, manufacturing dates and expiry dates.

This is meant to make sure the drug that has a short expiry is let out first so as to prevent expired stock losses.

- **Users to Transactions (1:M, MANDATORY PARTICIPATION)**: A user recorded in the USERS table can process multiple checkout transactions overtime but each individual transaction is tied to one specific user for purposes of accountability.
  The UserID foreign key in the TRANSACTIONS table tracks which staff member processed a given sale.

- **Transactions to Sale items (1:N, STRONG RELATIONSHIP)**: A single checkout transaction recorded in TRANSACTIONS can contain multiple purchased items in the SALES_ITEMS table.
  The TransactionID foreign key in SALES_ITEMS groups individual purchase rows under their parent transaction receipt.

- **Inventory Batches to Sale Items (1:M**): A specific stock batch from the INVENTORY_BATCHES can be sold across multiple customer transactions overtime until its quantity is depleted.
  The BatchID foreign key in SALES_ITEMS ensures that every sale item instantly deducts quantity from the correct batch and together with the transaction link, this acts as a junction between the transactions and inventory batches.

- **Categories to Drugs (ONE-TO-MANY):** One drug category can contain multiple generic medications but each drug belongs to one specific category.

- **Suppliers to Inventory Batches (ONE-TO-MANY):** One pharmaceutical supplier can deliver multiple inventory batches overtime but each individual batch comes from one specific supplier.